Name Reactions

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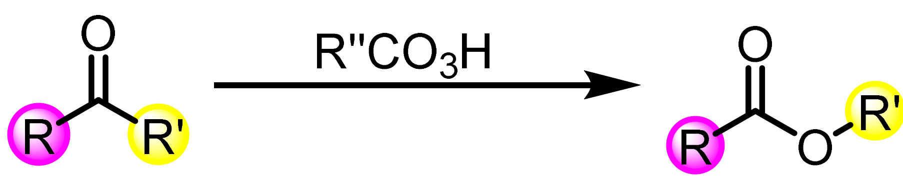
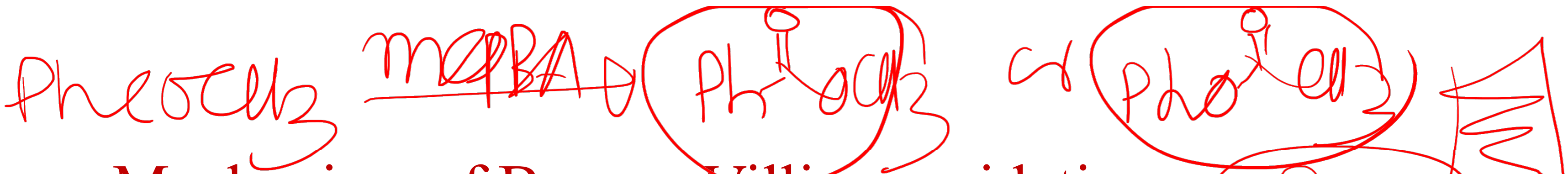
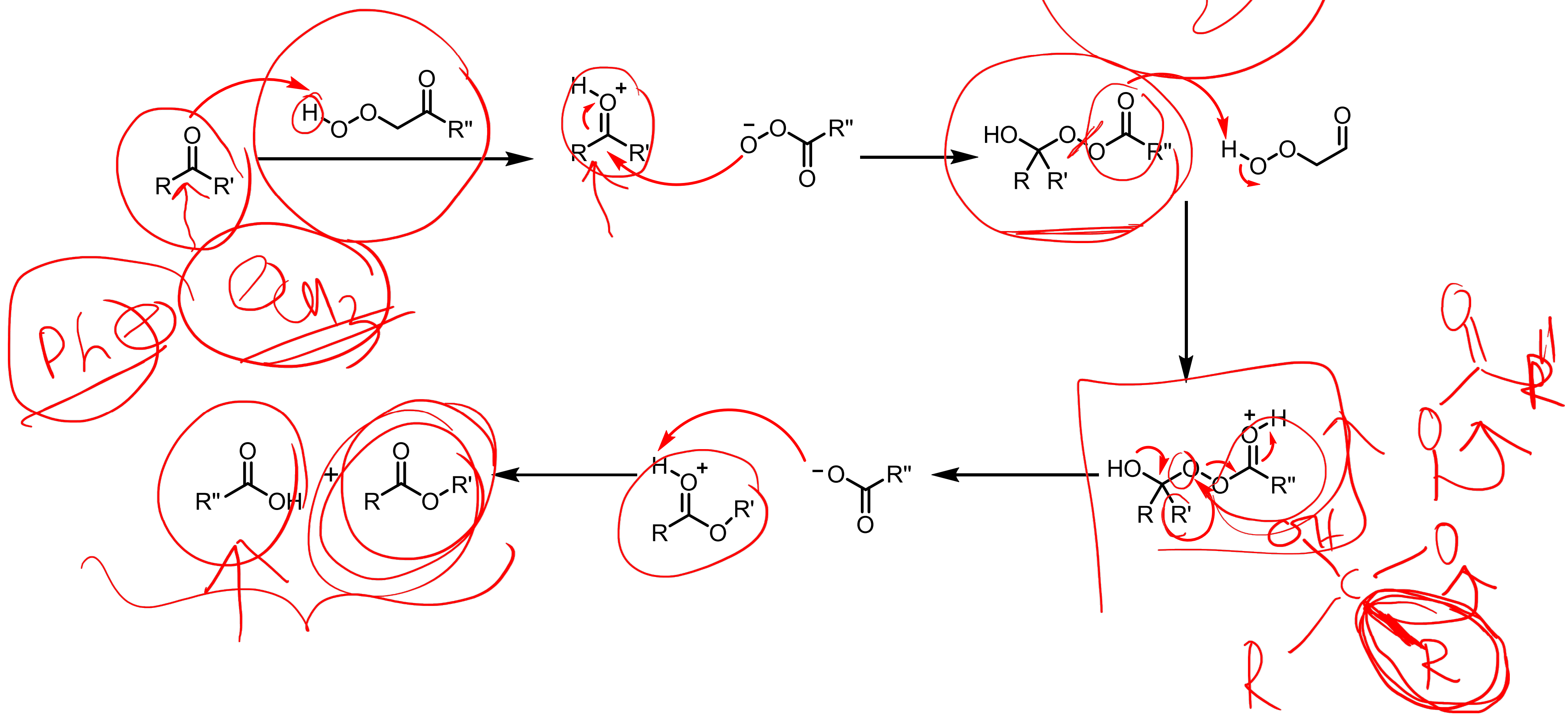
Baeyer–Villiger oxidation

- The Baeyer-Villiger Oxidation is the oxidative cleavage of a carbon-carbon bond adjacent to a carbonyl, and insertion of an oxygen atom which converts ketones to esters and cyclic ketones to lactones
- The Baeyer-Villiger can be carried out with peracids, such as *m*-CPBA, or with peroxide and a Lewis acid.

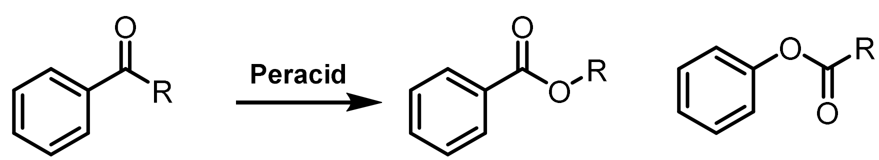




Mechanism of Baeyer-Villiger oxidation



- The reaction is regiospecific in nature and depends on the relative migration ability of the substituents attached to the carbonyl group.
- In unsymmetrical ketones that group migrates which is more electron releasing.
- Migratory aptitude of alkyl groups is in the order $3° > 2° > 1° > CH_3$
- Electron releasing substituents in the aryl group facilitate migration. The migratory order of aryl groups is *p*–anisyl > *p*–tolyl > phenyl > *p*–chlorophenyl > *p*–nitro-aryl.



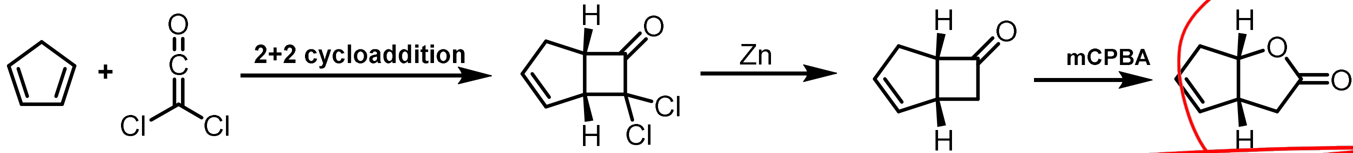
Via migration of R

Via migration of Ph

Examples:

mCPBA + $NaHCO_3$?

1.



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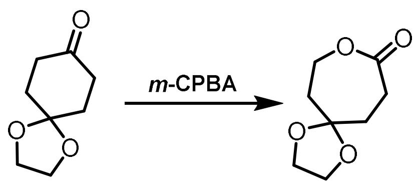

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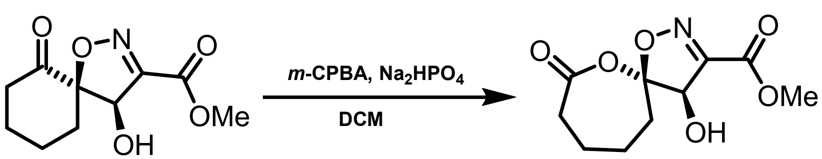

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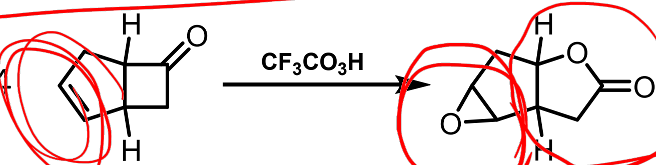

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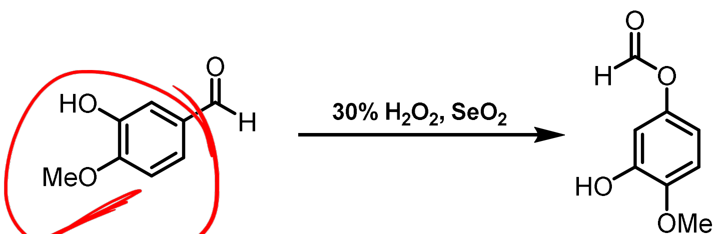
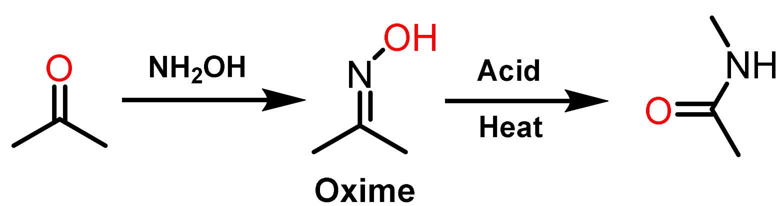


Beckman rearrangement

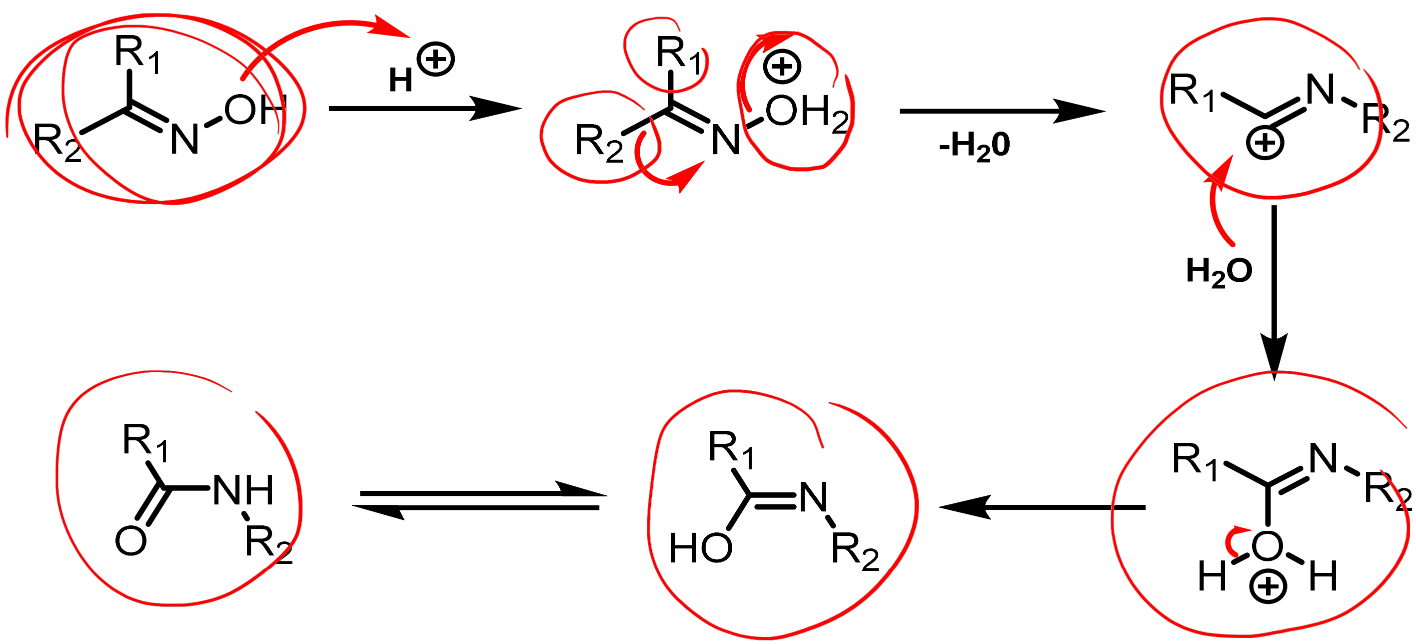
- The Beckmann rearrangement is a rearrangement of an oxime functional group to substituted amides.
- It is often catalysed by acid; however, other reagents have been known to promote the rearrangement. These include tosyl chloride, thionyl chloride, phosphorus pentachloride, etc



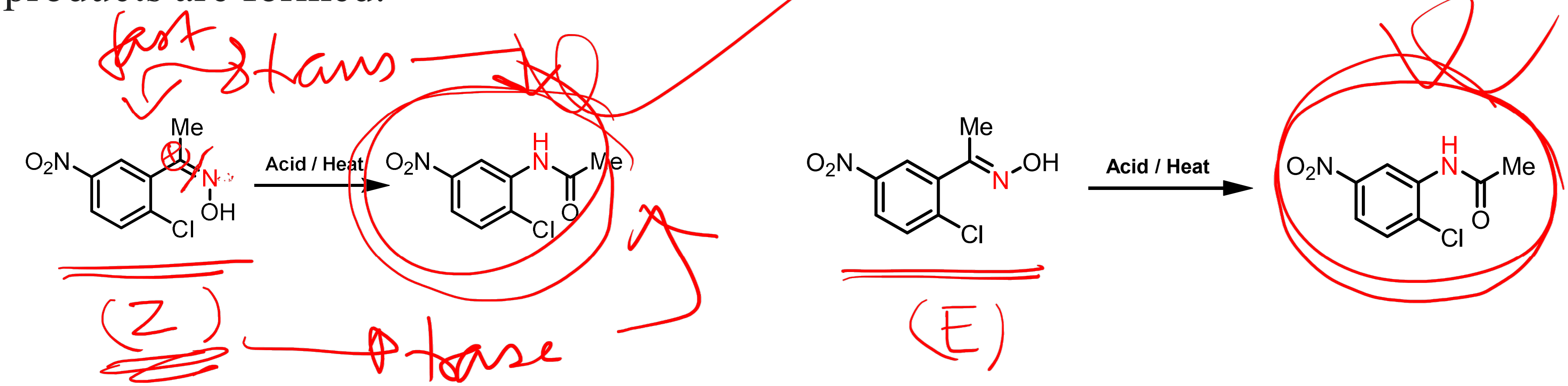
Mechanism:

Steps:

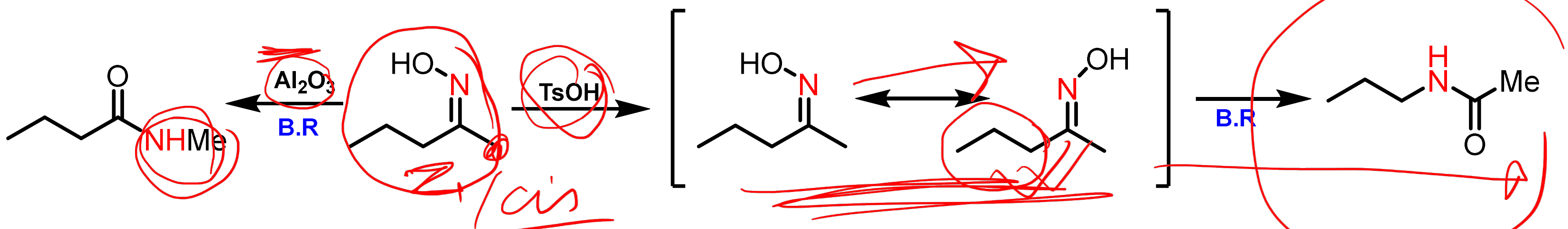
- The oxime oxygen is protonated by acid, This leads to formation of OH2+, which is a much better leaving group than OH
- The C-C bond breaks, migrating to the nitrogen, forming a new C-N bond and displacing the water as a leaving group
- This forms a free carbocation, which is then attacked by water
- The positively charged oxygen is deprotonated by base to give the tautomer of the amide.
- Then tautomerism occur to give final product.



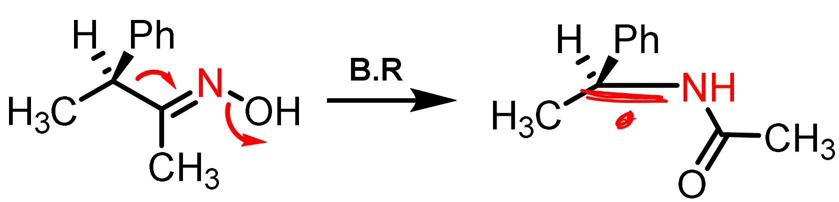
- The rearrangement occurs stereospecifically for ketoximes and N-chloro/N-fluoro imines, with the migrating group being anti-periplanar to the leaving group on the nitrogen.
- The rearrangement of aldoximes occurs with stereospecificity in the gas phase and without stereospecificity in the solution phase.
- Nitrone rearrangement also occurs without stereospecificity; the regioisomer formed has the amide nitrogen substituted with the group possessing the greatest migratory aptitude.
- Rearrangement is strictly intermolecular S_N2 in nature, hence no crossover products are formed.

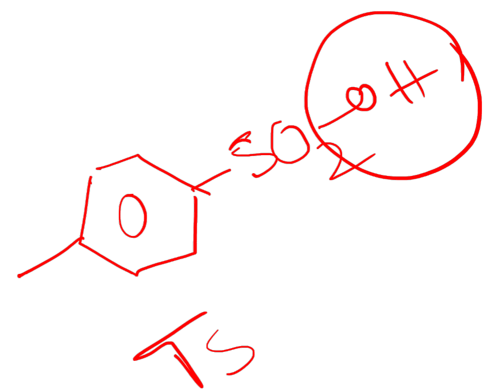


- Some reagents shows oxime isomerization faster than backmann rearrangement like TsOH, leading different product.

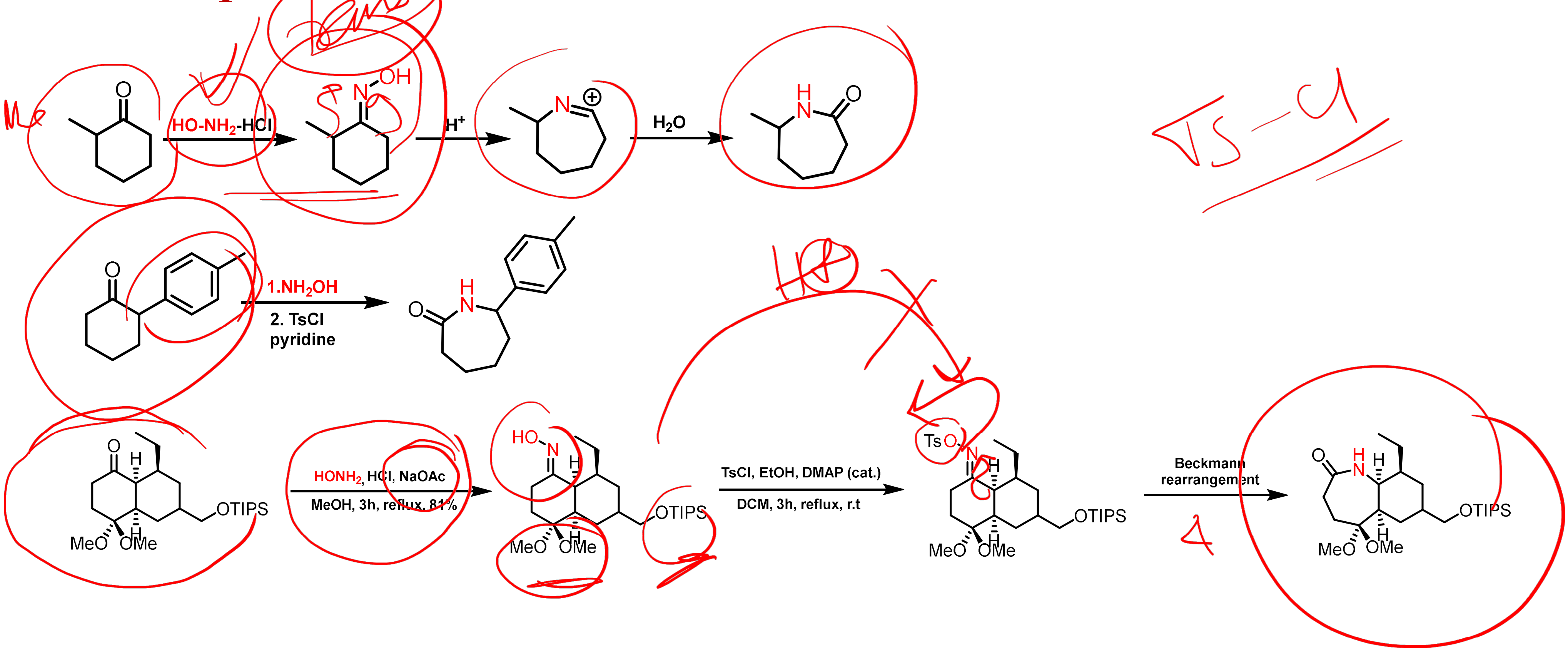


- If migrating group is chiral it will retain its stereochemistry during migration.



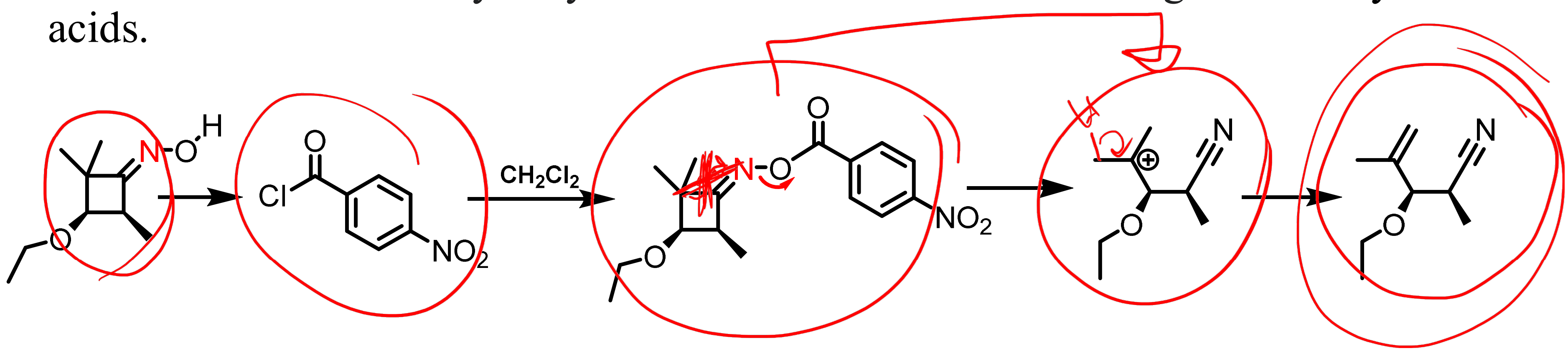


Examples:


HO-NH2-HCl
N-OH
H+
N
H2O
H
N
O
1.NH2OH
2. TsCl
pyridine
HONH2, HCl, NaOAc
MeOH, 3h, reflux, 81%
OTIPS
MeO
OMe
HO
TsCl, EtOH, DMAP (cat.)
DCM, 3h, reflux, r.t
TsO
Beckmann
rearrangement

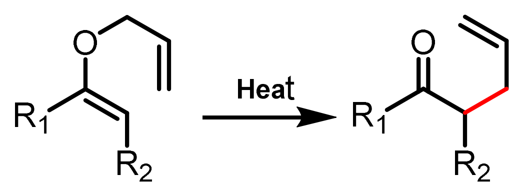
Beckmann fragmentation :

- The Beckmann fragmentation is another reaction that often competes with the rearrangement, though careful selection of promoting reagent and solvent conditions can favor the formation of one over the other, sometimes giving almost exclusively one product.
- When the group α to the oxime is capable of stabilizing carbocation formation, the fragmentation becomes a viable reaction pathway.
- The reaction generates a nitrile and a carbocation, which is quickly intercepted to form a variety of products
- The nitrile can also be hydrolyzed under reaction conditions to give carboxylic acids.

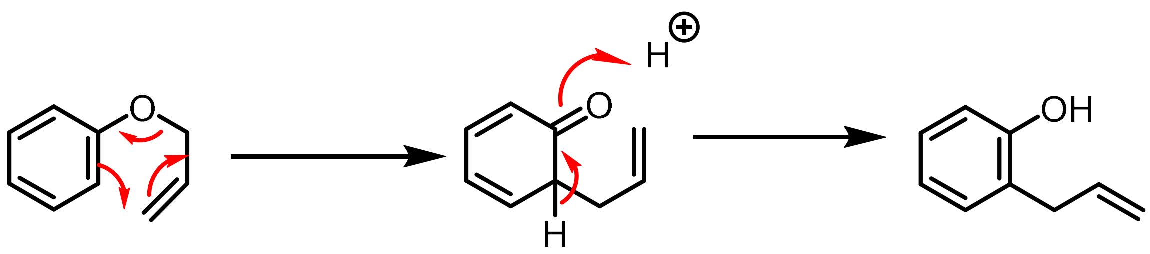


Claisen Rearrangement

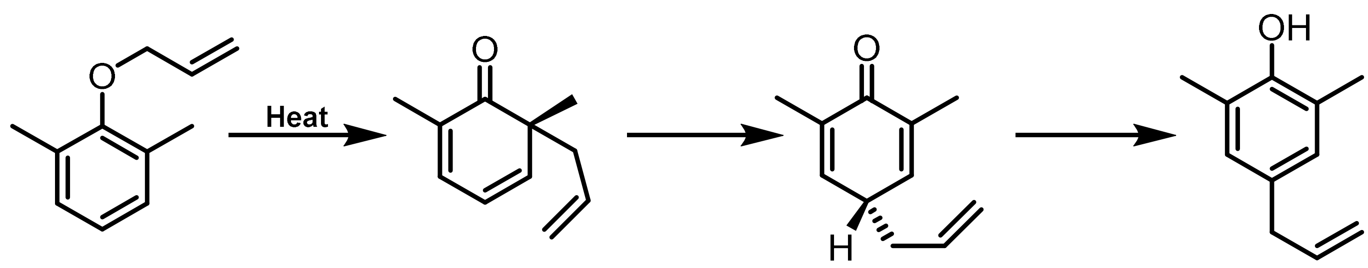
- Claisen rearrangement reaction is [3+3] sigmatropic rearrangement reaction.
- Rearrangement of allyl vinyl ethers to the corresponding γ,δ-unsaturated carbony compounds.



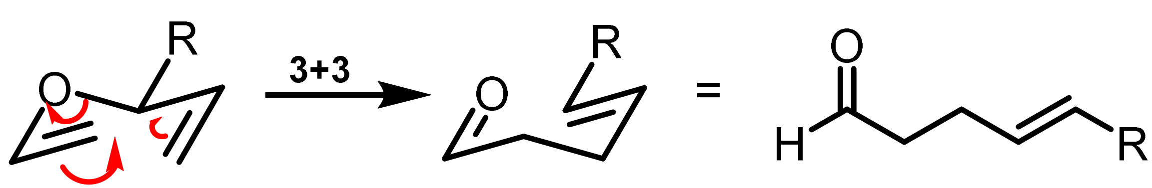
- This is a one-step mechanism without ionic intermediates or any charges.



- Allyl group also appears to attack the *para* position on the ring.

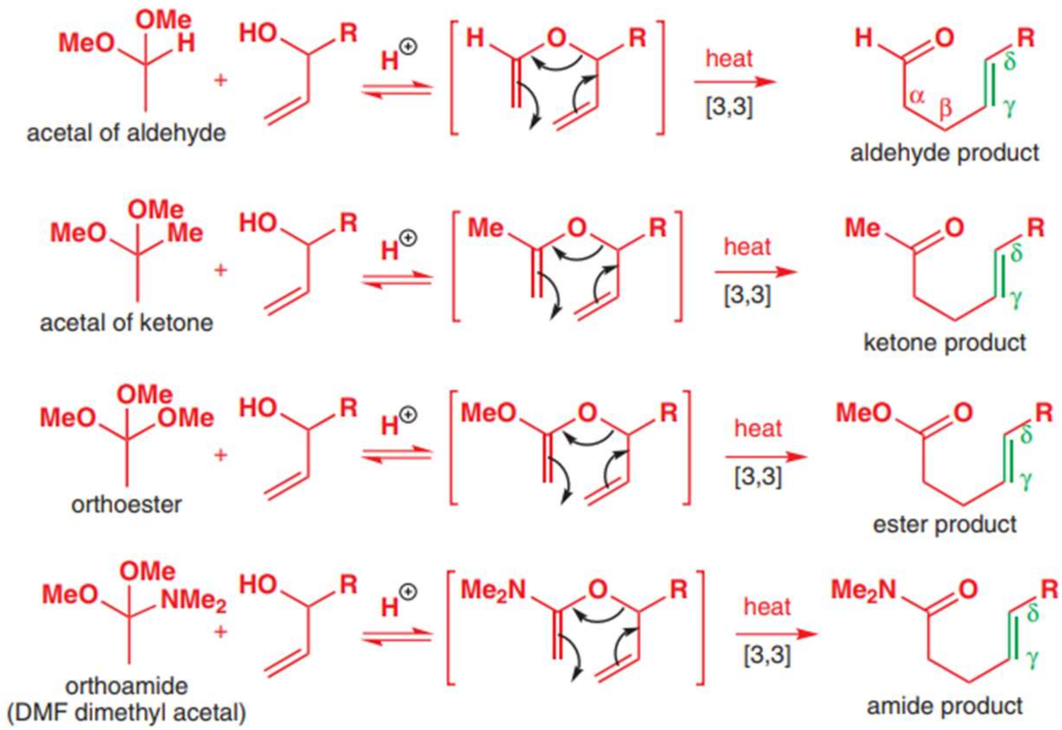


- Stereochemistry may arise if there is a substituent on the saturated carbon atom next to the oxygen atom.
- The resulting double bond strongly favors the trans *(E)* geometry
- Because substituent prefers an equatorial position on the chair transition state.

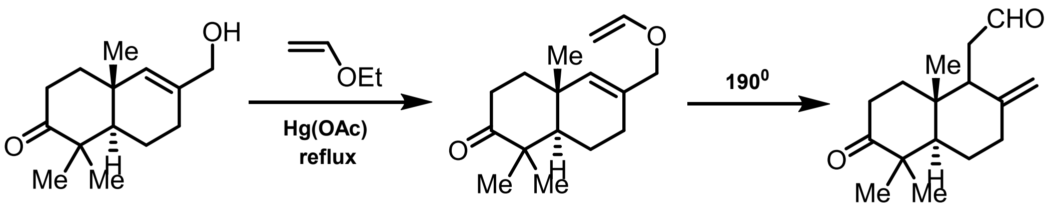


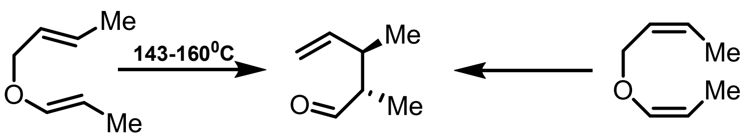
Significance:

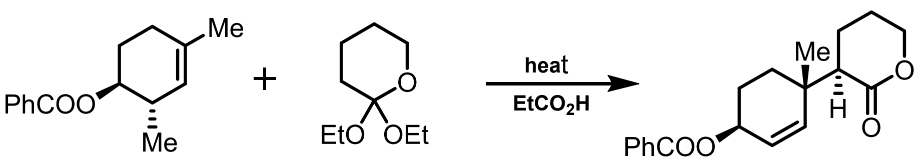
- General synthesis of γ,δ-unsaturated carbonyl compounds.

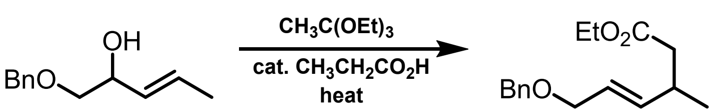


Examples:


OH
Me
Me Me
OEt
Hg(OAc)
reflux
190⁰
CHO

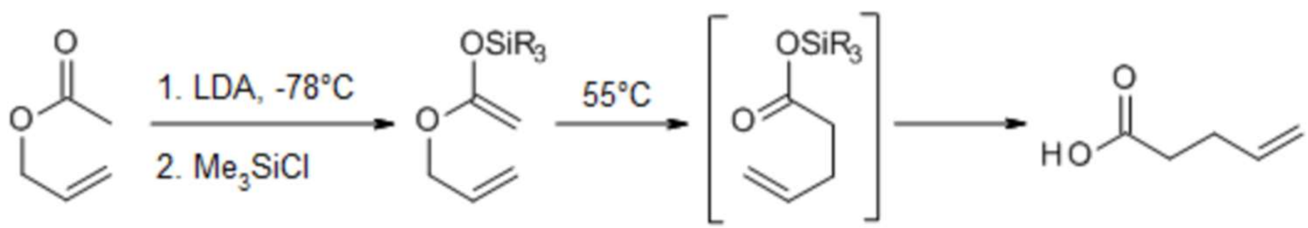

Me
143-160⁰C
Me


PhCOO
Me
EtO OEt
heat
EtCO2H

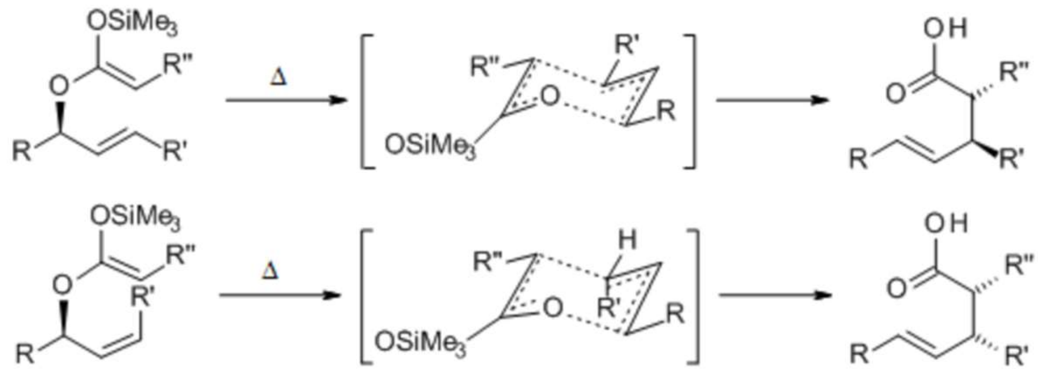

BnO
OH
CH3C(OEt)3
cat. CH3CH2CO2H
heat
EtO2C

Ireland–Claisen rearrangement

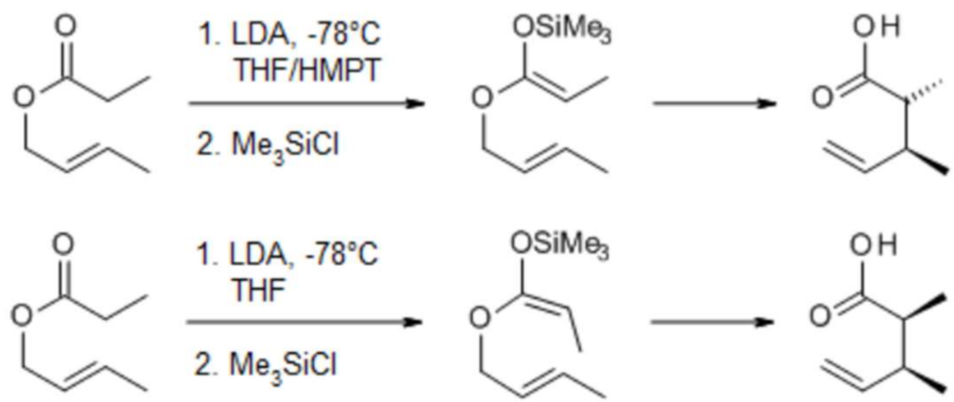
- Variant of the Claisen Rearrangement.
- The Ireland–Claisen rearrangement is the reaction of an allylic carboxylate with a strong base
- It give a γ,δ-unsaturated carboxylic acid.
- The rearrangement proceeds via silylketene acetal.



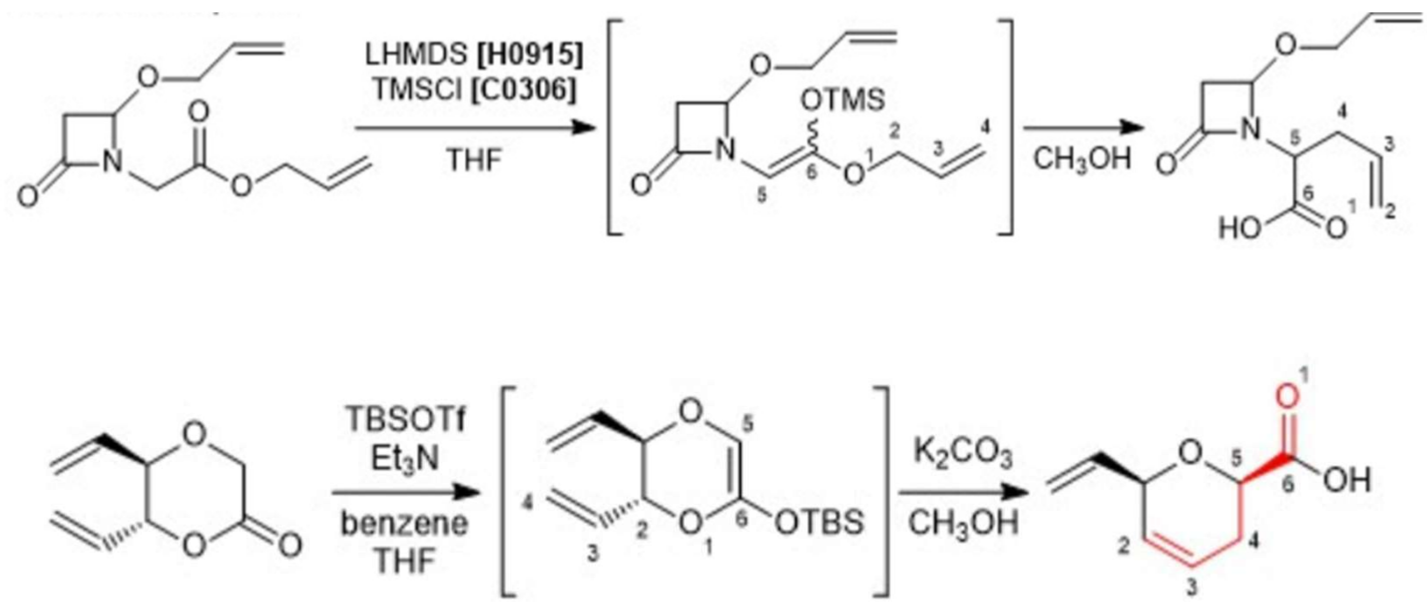
- Ireland modification proceeds with a high degree of stereoselectivity:



- An advantage of the Ireland-Claisen Rearrangement is the option of controlling the enolate geometry through the judicious choice of solvent.



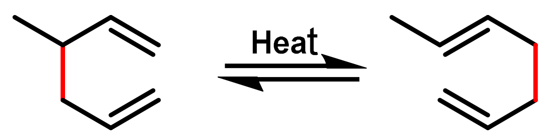
Examples:



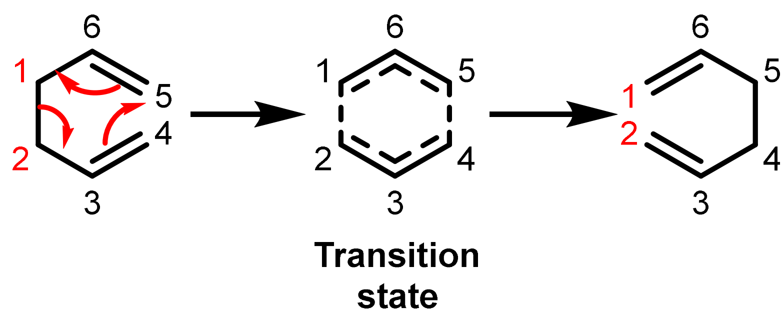
LHMDS [H0915]
TMSCl [C0306]
THF
OTMS
CH3OH
HO
TBSOTf
Et3N
benzene
THF
OTBS
K2CO3
CH3OH
OH

Cope rearrangement reaction

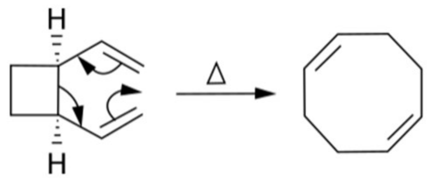
The Cope rearrangement is a [3,3]-sigmatropic rearrangement of 1,5-dienes.

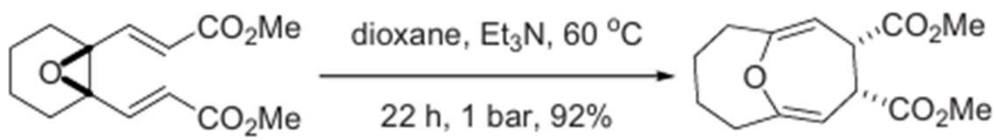


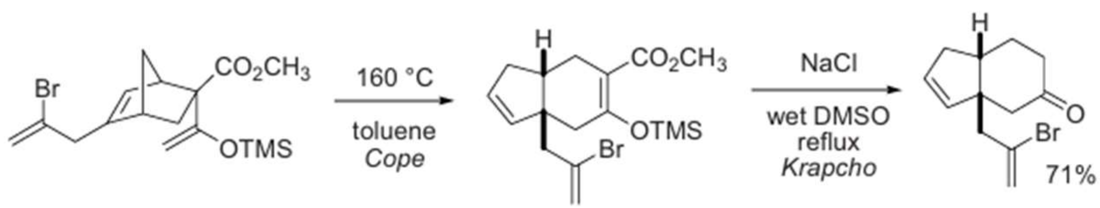
The Cope rearrangement occurs in a single step with a concerted transition state.



Examples:

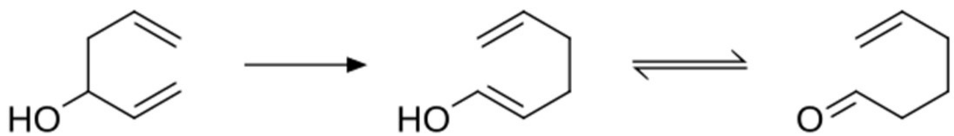

H
H
Δ


CO2Me
O
CO2Me
dioxane, Et3N, 60 °C
22 h, 1 bar, 92%
O
CO2Me
CO2Me

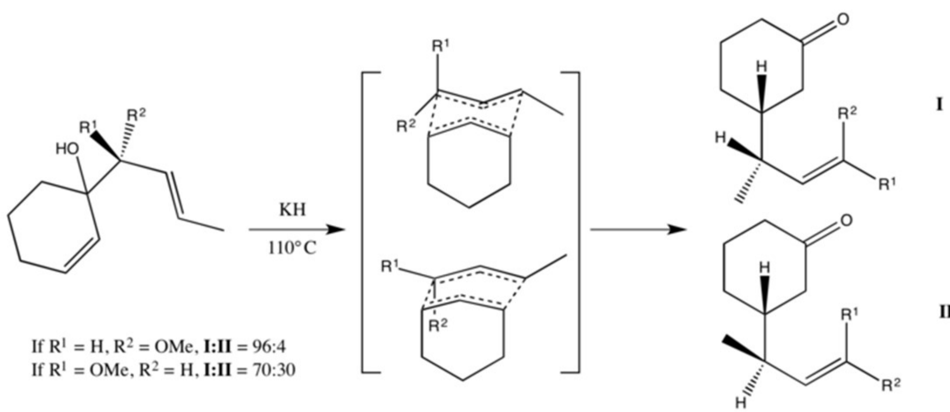

Br
CO2CH3
OTMS
160 °C
toluene
Cope
H
CO2CH3
OTMS
Br
NaCl
wet DMSO
reflux
Krapcho
H
O
Br
71%

Oxy cope rearrangement

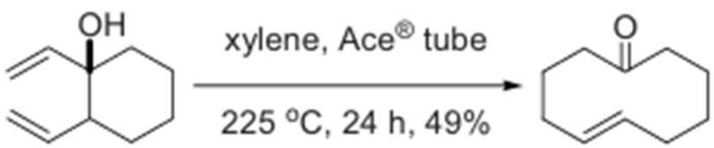
- In the oxy-Cope rearrangement, a hydroxyl group is added at C3 forming an enone after keto-enol tautomerism of the intermediate enol.

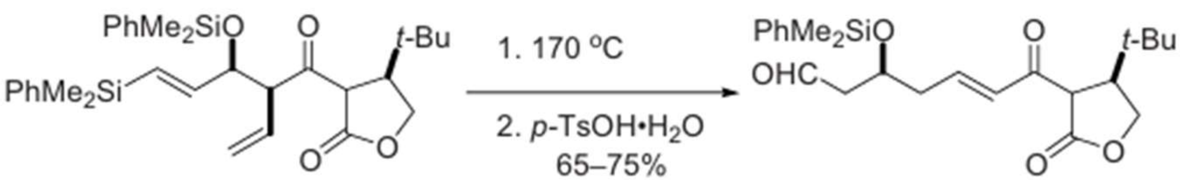


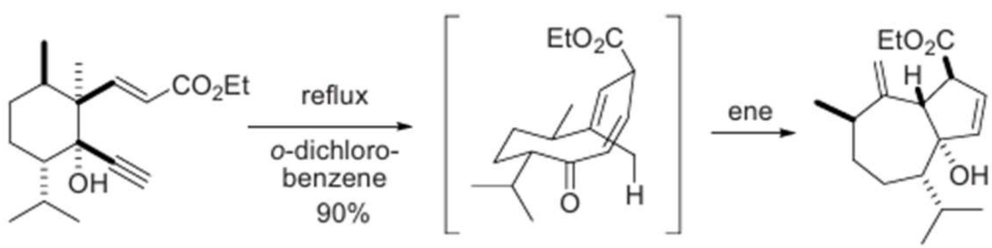
Mechanism:



Examples:


OH
xylene, Ace® tube
225 °C, 24 h, 49%
O


PhMe2SiO
O
t-Bu
PhMe2Si
O
O
1. 170 °C
2. p-TsOH•H2O
65–75%
PhMe2SiO
OHC
O
t-Bu
O
O


CO2Et
OH
reflux
o-dichloro-
benzene
90%
EtO2C
O
H
ene
EtO2C
H
OH